

Lab Report – 1

Course: EEE 3400 (B): Numerical Techniques Laboratory

Semester: Summer 2026

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Solve of RLC Series Circuit:

Matlab Code:

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clear all; clc
%Given
R = 220;
C = 0.6e-6;
L = 0.5;
Z_targetValue = 400;
% defining RLC impedance function f(w)=0
f=@(w) sqrt(R^2 +(w*L -1./(w*C)).^2)-Z_targetValue;
%answer of 1
f_500 =f(500);
f_1000= f(1000);
fprintf('f(500)= %f, f(1000) = %f\n',f_500,f_1000);
fprintf('Answer:No, it will not converge because both values are positive.\n');
fprintf('There is no sign change,so f(500)*f(1000)>0.\n\n');
%answer of 2
xl =1200;
xu= 1700;
fprintf('Chosen Interval:[%d, %d]\n',xl, xu);
fprintf('f(1200) = %f, f(1700) = %f\n',f(xl),f(xu));
fprintf('Answer: This is valid because f(1200) is negative and f(1700) is positive.\n');
fprintf('Signs are opposite, so f(xl)*f(xu)< 0.\n\n');
%answer of 3
%formula: n>=log2(initial_width/ tolerance)
n_steps = ceil(log2((xu - xl)/ 10^-4));
fprintf('Answer: Total steps required = %d steps.\n\n',n_steps);
%Answer of 4
%Step 1
xr1 = (xl + xu) / 2;
fprintf('Step 1 -> xl: %d, xu: %d, xr: %.2f, f(xr): %.2f\n',xl,xu,xr1,f(xr1));
if f(xl)*f(xr1) < 0
xu=xr1;
else
xl=xr1;
end
%Step2
xr2 = (xl + xu) / 2;
fprintf('Step 2 -> xl: %d, xu: %d, xr: %.2f, f(xr): %.2f\n', xl, xu, xr2, f(xr2));
if f(xl)*f(xr2) < 0
xu=xr2;
else
xl=xr2;
end
%Step3
xr3 = (xl + xu) / 2;
fprintf('Step 3 -> xl: %d, xu: %d, xr: %.2f, f(xr): %.2f\n', xl, xu, xr3, f(xr3));
if f(xl)*f(xr3) < 0
xu=xr3;
else
xl=xr3;
end
fprintf('\n');
fprintf('--- Finding final accurate root using remaining iterations\n');
es = 10^-4; %target error
ea = 100; %initial high error to start loop
xr = xr3;% start from step 3 result
while ea > es
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xr_old =xr;
xr =(xl+ xu) / 2; % step 2 from sheet
if xr_old~=0
ea =abs((xr - xr_old) / xr) * 100; % error calculation from step 4
end
% step3 from sheet
if f(xl) *f(xr) < 0
xu = xr;
elseif f(xl)* f(xr) > 0
xl=xr;
else
break;
end
end
fprintf('Final convergent root w=%f rad/s\n',xr);

```

Command Window Output:

$f(500) = 2691.172018$, $f(1000) = 787.228331$
 Answer: No, it will not converge because both values are positive.
 There is no sign change, so $f(500)*f(1000) > 0$.

Chosen Interval: [1200, 1700]
 $f(1200) = 418.990646$, $f(1700) = -144.261629$
 Answer: This is valid because $f(1200)$ is negative and $f(1700)$ is positive.
 Signs are opposite, so $f(xl)*f(xu) < 0$.

Answer: Total steps required = 23 steps.

Step 1 -> xl: 1200, xu: 1700, xr: 1450.00, $f(xr)$: 78.06
 Step 2 -> xl: 1450, xu: 1700, xr: 1575.00, $f(xr)$: -51.17
 Step 3 -> xl: 1450, xu: 1575, xr: 1512.50, $f(xr)$: 9.75

--- Finding final accurate root using remaining iterations

Final convergent root $w = 1521.988106$ rad/s

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```

MATLAB Drive/RLSeriesCircuitP1.m
1 clear all; clc
2
3 %Given
4 R = 220;
5 C = 0.6e-6;
6 L = 0.5;
7 Z_targetValue = 400;
8
9 % defining RLC impedance function f(w)=0
10 f=@(w) sqrt(R^2 +(w*L -1./(w*C)).^2)-Z_targetValue;
11
12 %answer of 1
13 f_500=f(500);
14 f_1000=f(1000);
15 fprintf('f(500)= %f, f(1000) = %f\n',f_500,f_1000);
16 fprintf('Answer:No, it will not converge because both values are positive.\n');
17 fprintf('There is no sign change,so f(500)*f(1000)>0.\n\n');
18
19 %answer of 2
20 xl =1200;
21 xu= 1700;
22 fprintf('Chosen Interval:[%d, %d]\n',xl, xu);
23 fprintf('f(1200) = %f, f(1700) = %f\n',f(xl),f(xu));
24
25 fprintf('Answer: This is valid because f(1200) is negative and f(1700) is positive.\n');
26 fprintf('Signs are opposite, so f(xl)*f(xu)< 0.\n\n');
27
28 %answer of 3
29 %formula: n>=log2(initial_width/ tolerance)
30 n_steps = ceil(log2((xu - xl)/ 10^-4));
31 fprintf('Answer: Total steps required = %d steps.\n\n',n_steps);
32
33 %Answer of 4

```

Command Window

```

f(500)= 2691.172018, f(1000) = 787.228331
Answer:No, it will not converge because both values are positive.
There is no sign change,so f(500)*f(1000)>0.

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f(1200) = 418.990646, f(1700) = -144.261629
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Answer: Total steps required = 23 steps.

Step 1 -> xl: 1200, xu: 1700, xr: 1450.00, f(xr): 78.06
Step 2 -> xl: 1450, xu: 1700, xr: 1575.00, f(xr): -51.17
Step 3 -> xl: 1450, xu: 1575, xr: 1512.50, f(xr): 9.75

--- Finding final accurate root using remaining iterations
Final convergent root w=1521.988106 rad/s
>>

```